

SO(3) Equivariant Neural Networks

Lecture 3 Solutions

How Geometric Types Combine

Solutions to checkpoints, discussions, and notebook exercises

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1 Solutions to lecture questions

Question 3.1

The triangle rule gives

$$0 \otimes 2 = 2, \quad 1 \otimes 2 = 1 \oplus 2 \oplus 3, \quad 2 \otimes 2 = 0 \oplus 1 \oplus 2 \oplus 3 \oplus 4.$$

Dimension checks are $1 \cdot 5 = 5$; $3 \cdot 5 = 3 + 5 + 7 = 15$; and $5 \cdot 5 = 1 + 3 + 5 + 7 + 9 = 25$.

Question 3.2

For two vectors, the $L = 0$ component is proportional to the dot product, the $L = 1$ component is proportional to the cross product, and the $L = 2$ component is the symmetric-traceless part of the outer product. Their dimensions are 1, 3, and 5.

Question 3.3

The uncoupled product basis labels components by (m_1, m_2) . The coupled basis labels them by (L, M) . CG coefficients are entries of the unitary transformation between these coordinate systems, organized by total output angular momentum.

Question 3.4

CG coefficients are fixed representation data and are loaded from SymPy. Learnable flexibility enters through channel multiplicities and weight matrices that mix the channel-pair outputs of each allowed CG path. Learned weights never replace the angular coupling table.

Question 3.5

Route one rotates both inputs with $D^{(\ell_1)}$ and $D^{(\ell_2)}$, then applies the CG product. Route two applies the CG product first and rotates the resulting L multiplet with $D^{(L)}$. The intertwining identity says the routes agree.

Question 3.6

Correct shape only makes the contraction syntactically possible. Equivariance imposes additional linear constraints on the tensor entries. A random tensor almost surely fails those constraints and therefore does not commute with the representation matrices.

Question 3.7

Degree four. Track the inputs through the network: the two $\ell = 1$ inputs are degree one. Inside the first layer, a channel-mixing path (lin, ℓ) multiplies by a learned matrix and leaves the degree at one, while a coupling path $\ell_1 \otimes \ell_2 \rightarrow \ell_3$ multiplies two degree-one features and returns degree two. So the hidden features carry degree one on the linear paths and degree two on the coupling paths. The second layer does the same thing again. Its coupling paths multiply two hidden features, and the highest available combination is degree two times degree two, giving degree four. That is the maximum, and it is achieved by taking a coupling path in the first layer for *both* arguments of a coupling path in the second: $1 \otimes 1 \rightarrow \ell_a$ and $1 \otimes 1 \rightarrow \ell_b$ in layer one, then $\ell_a \otimes \ell_b \rightarrow 1$ in layer two. The paths that cannot raise the degree are the channel-mixing ones, (lin, ℓ) . They are linear maps

on the channel index at fixed ℓ , so they preserve whatever degree arrives.

One caution: the network output is a sum over *all* paths, so it is an inhomogeneous polynomial containing degrees 1 through 4, not a pure degree-four form. That is why the log-log slope in the Lecture 4 notebook *climbs* from 1 to 4 rather than sitting at 4: at small λ the lowest degree dominates, at large λ the highest does.

2 Notebook reference checks

The direct intertwiner residual is

```

xr = D_sph(l1, R) @ x
yr = D_sph(l2, R) @ y
lhs = cg_product(xr, yr, l1, l2, L)
rhs = D_sph(L, R) @ cg_product(x, y, l1, l2, L)
residual = np.max(np.abs(lhs-rhs))

```

For library CGCs consistent with the basis convention, expect roughly 10^{-15} . A normalized random tensor generally produces a residual many orders of magnitude larger.

For physical vectors u, v , convert them with `cart_to_sph`. With the phase convention used in the series the constants are exact rather than approximate:

$$z^{(0)} = \frac{u \cdot v}{\sqrt{3}}, \quad z^{(1)}|_{\text{Cartesian}} = \frac{u \times v}{\sqrt{2}},$$

and the $L = 2$ product has five components. Both identities hold to machine precision, so the notebook checks them as residuals rather than estimating a scale factor.